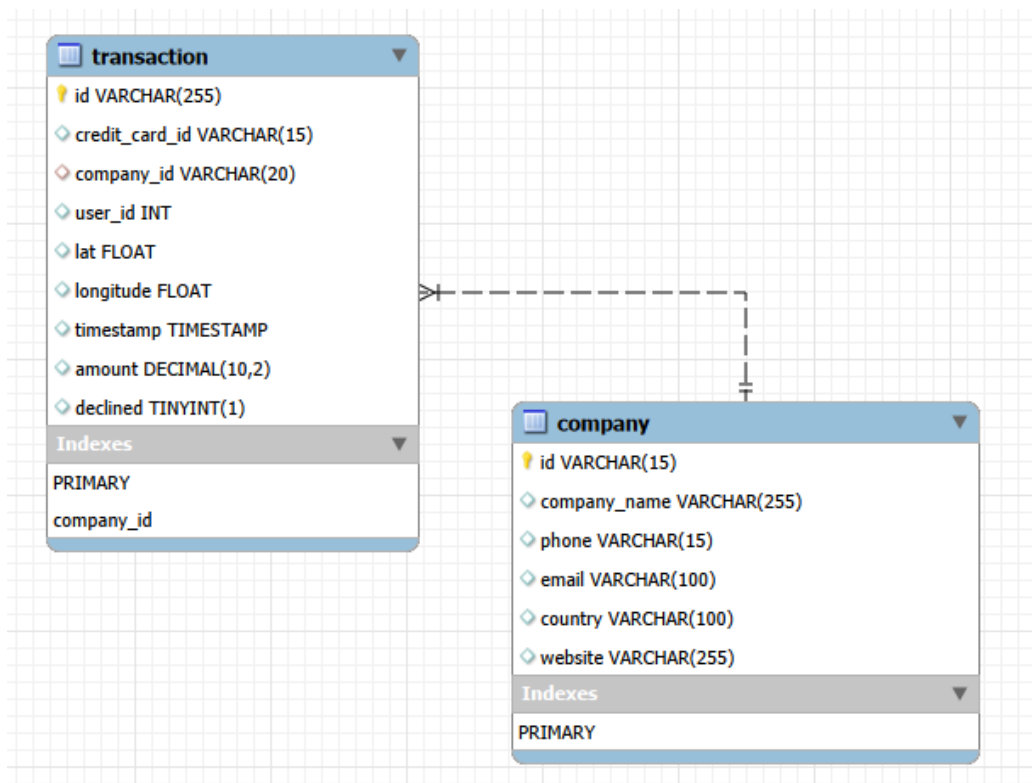


EJERCICIOS - Tasca S2.01. Nocions bàsiques SQL

NIVELL 1

EJERCICIO 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



La base de datos está formada por dos tablas: **transaction** y **company**. Los elementos característicos de cada tabla son:

- **Transaction:**
 - 9 campos con 1 clave primaria (id) en formato VARCHAR de longitud 255 y clave foranea (company_id) de la tabla company.
 - Tiene 3 campos de tipo VARCHAR, 1 de tipo INT, 2 de tipo FLOAT, 1 de tipo DECIMAL, 1 de tipo TIMESTAMP y 1 TINYTINT.
 - Existe un indice con la clave primaria y la clave foranea “company_id”
- **Company:**
 - 6 campos con 1 clave primaria (id) en formato VARCHAR de longitud 15.
 - Todos los campos son de tipo VARCHAR.
 - Existe un indice con la clave primaria.

El tipo de relación que las une es de **1:N** a través del campo **company_id**. Es la clave primaria (PK) de company y la clave foranea (FK) de transaction.

EJERCICIO 2

Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes

```
10 --- Llistat dels països que estan generant vendes.  
11 • SELECT distinct(c.country) as "países_vendas"  
12 FROM company as c  
13 JOIN transaction as t on c.id = t.company_id  
14 WHERE t.declined <> 0  
15 GROUP BY c.country  
16 ORDER BY c.country;  
17
```

Result Grid				
Filter Rows: Export: Wrap Cell Content:				
países_vendas				
▶ Australia				
Belgium				
Canada				
China				
France				
Germany				
Ireland				

Result 6				
Output				
Action Output				
#	Time	Action	Message	
✓ 14	15:11:48	CREATE TABLE IF NOT EXISTS product_transaction SELECT t.id as transaction_id, p.id ...	253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0	
✓ 15	15:12:30	SELECT distinct(product_id) as product_id, count(product_id) as 'h_vendas' FROM product...	100 row(s) returned	
✓ 16	15:14:10	SELECT * FROM transactions.transaction	100001 row(s) returned	
✓ 17	15:15:03	SELECT distinct(c.country) as "países_vendas" FROM company as c JOIN transaction as ...	15 row(s) returned	

- Des de quants països es generen les vendes.

```
19 • SELECT count(distinct(c.country)) as "n_países"  
20 FROM company as c  
21 JOIN transaction as t on c.id = t.company_id  
22 WHERE t.declined <> 0  
23 ORDER BY c.country;  
24  
25 --- Identifica la companyia amb la mitjana més gran de vendes.  
26 • SELECT c.company_name, avg(count) as "media_vendas"
```

Result Grid				
Filter Rows: Export: Wrap Cell Content:				
n_países				
▶ 15				

Result 7				
Output				
Action Output				
#	Time	Action	Message	
✓ 15	15:12:30	SELECT distinct(product_id) as product_id, count(product_id) as 'h_vendas' FROM product...	100 row(s) returned	
✓ 16	15:14:10	SELECT * FROM transactions.transaction	100001 row(s) returned	
✓ 17	15:15:03	SELECT distinct(c.country) as "países_vendas" FROM company as c JOIN transaction as ...	15 row(s) returned	
✓ 18	15:16:22	SELECT count(distinct(c.country)) as "n_países" FROM company as c JOIN transaction a...	1 row(s) returned	

- Identifica la companyia amb la mitjana més gran de vendes.

```

26 • SELECT c.company_name, round(avg(amount),2) as "media_ventas"
27 FROM company as c
28 JOIN transaction as t on c.id = t.company_id
29 WHERE t.amount > 0
30 GROUP BY c.company_name
31 ORDER BY media_ventas desc
32 LIMIT 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
company_name	media_ventas			
Ac Fermentum Incorporated	284.87			

Result 13 x

Output				
Action Output				
#	Time	Action	Message	
✓ 21	15:17:19	SELECT c.company_name, round(avg(amount),2) as "media_ventas" FROM company as ...	1 row(s) returned	
✓ 22	15:18:17	SELECT c.company_name, avg(amount) as "media_ventas" FROM company as c JOIN tr...	101 row(s) returned	
✓ 23	15:18:29	SELECT c.company_name, round(avg(amount),2) as "media_ventas" FROM company as ...	101 row(s) returned	
✓ 24	15:18:39	SELECT c.company_name, round(avg(amount),2) as "media_ventas" FROM company as ...	1 row(s) returned	

EJERCICIO 3

Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya

ejercicios_noel*

Don't Limit

36

-- Mostra totes les transaccions realitzades per empreses d'Alemanya.

37

• select subq1.*

38

from (

39

SELECT t.*, c.country

40

FROM company as c,

41

transaction as t

42

WHERE c.id = t.company_id) as subq1

43

WHERE country = "Germany";

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined	country
▶	00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0	Germany
	0013C1B6-3884-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0	Germany
	00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0	Germany
	00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0	Germany
	005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0	Germany
	00687139-48B2-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0	Germany
	0074F4DD-32F1-4827-8758-55896314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0	Germany
	00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-6797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0	Germany

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 1	17:41:40	select subq1.* from (SELECT t.*, c.country FROM company as c, transaction as t...	13291 row(s) returned

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

ejercicios_noel

```

44
45 --- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.
46 • SELECT distinct(c.company_name) as "company_name"
47 FROM company as c,
48 transaction as t
49 WHERE c.id = t.company_id
50 and t.amount >
51 (SELECT avg(t.amount)
52 FROM transaction as t);

```

Result Grid

company_name
Eget Tincidunt Dui Institute
Neque Tellus Imperdiet Corp.
Aliquam Iaculis Lacus Corp.
Lorem Ipsum Dolor Corp.
Nulla Integer Vulputate Corp.
Auctor Mauris Corp.

Result 17

Output

Action Output

#	Time	Action	Message
1	18:01:49	SELECT distinct(c.company_name) as "company_name" FROM company as c, tran...	100 row(s) returned

- Eliminaren del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```

57 --- Eliminaren del sistema les empreses que no tenen transaccions registrades, entrega el llistat daquestes empreses.
58 • SELECT c.id, c.company_name
59 FROM company as c
60 WHERE c.id NOT IN (
61 SELECT t.company_id
62 FROM transaction as t
63 WHERE c.id = t.company_id);

```

Result Grid

id	company_name
NULL	NULL

company 19

Output

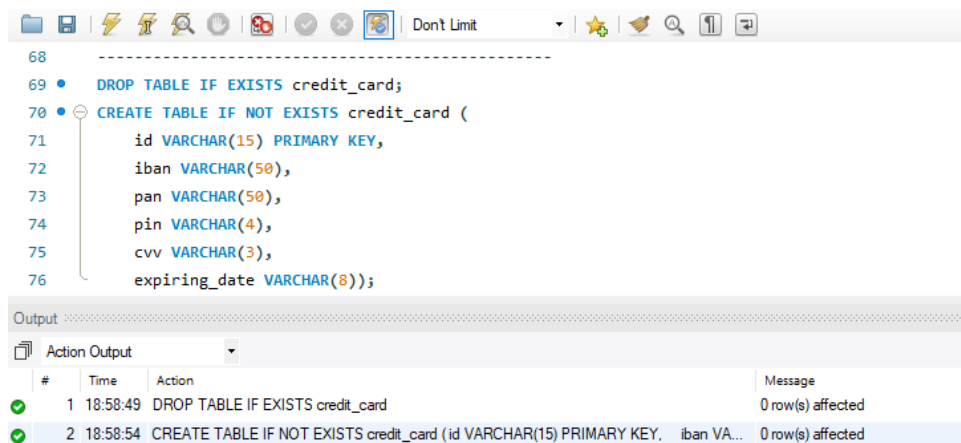
Action Output

#	Time	Action	Message
1	15:38:12	SELECT c.id, c.company_name FROM company as c WHERE c.id NOT IN (SELECT t.co...	0 row(s) returned

No hay empresas que no tengan transacciones sin registrar.

EJERCICIO 4

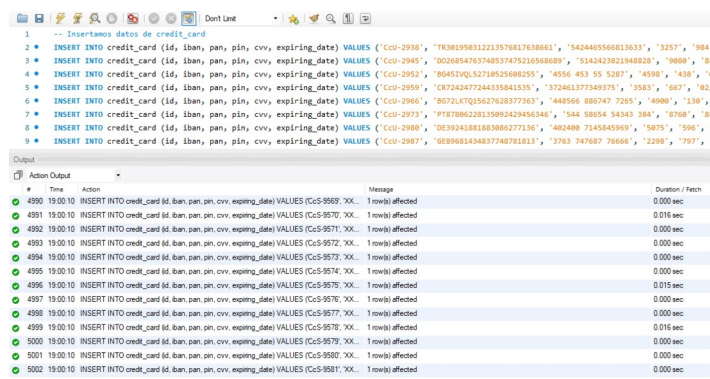
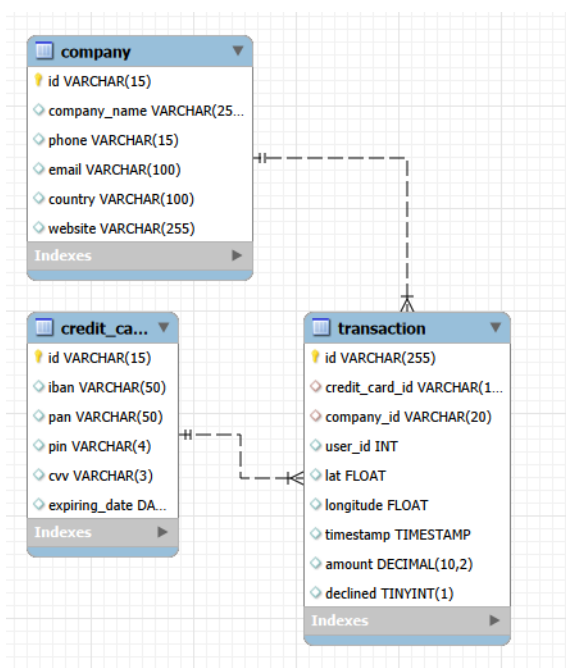
La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.



```
68 -----
69 • DROP TABLE IF EXISTS credit_card;
70 • CREATE TABLE IF NOT EXISTS credit_card (
71     id VARCHAR(15) PRIMARY KEY,
72     iban VARCHAR(50),
73     pan VARCHAR(50),
74     pin VARCHAR(4),
75     cvv VARCHAR(3),
76     expiring_date VARCHAR(8));
```

Output

#	Time	Action	Message
1	18:58:49	DROP TABLE IF EXISTS credit_card	0 row(s) affected
2	18:58:54	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(15) PRIMARY KEY, iban VA...	0 row(s) affected



```
1 -- Insertamos datos de credit_card
2 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2938', 'TR301950312213570817638661', '5424405566813633', '3257', '984', '1990-01-01');
3 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2945', 'D026854763748537475216568689', '5142423821948828', '9888', '88', '1990-01-01');
4 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2952', '80451VQL52718525688255', '4956 453 55 5287', '4588', '438', '8', '1990-01-01');
5 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2959', 'C07524277244835841535', '37246137746575', '3583', '667', '82', '1990-01-01');
6 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2966', '9672147015627628377963', '448566 685747 7265', '4486', '138', '1', '1990-01-01');
7 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2973', 'PT8788621815892429456346', '544 5854 54343 384', '8788', '88', '1990-01-01');
8 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2980', 'DE39241881883886277136', '482480 7145845969', '5875', '596', '1', '1990-01-01');
9 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-2987', 'GE8968143837748781813', '3763 747687 76666', '2296', '797', '1', '1990-01-01');
```

Output

#	Time	Action	Message	Duration / Fetch
4990	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9569, XX...	1 row(s) affected	0.000 sec
4991	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9570, XX...	1 row(s) affected	0.016 sec
4992	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9571, XX...	1 row(s) affected	0.000 sec
4993	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9572, XX...	1 row(s) affected	0.000 sec
4994	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9573, XX...	1 row(s) affected	0.000 sec
4995	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9574, XX...	1 row(s) affected	0.000 sec
4996	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9575, XX...	1 row(s) affected	0.015 sec
4997	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9576, XX...	1 row(s) affected	0.000 sec
4998	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9577, XX...	1 row(s) affected	0.000 sec
4999	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9578, XX...	1 row(s) affected	0.016 sec
5000	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9579, XX...	1 row(s) affected	0.000 sec
5001	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9580, XX...	1 row(s) affected	0.016 sec
5002	19:00:10	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CCU-9581, XX...	1 row(s) affected	0.000 sec

```
UPDATE credit_card
SET expiring_date = STR_TO_DATE(expiring_date, '%m/%d/%y')
WHERE id IS NOT NULL;

ALTER TABLE credit_card
MODIFY COLUMN expiring_date DATE;
SET FOREIGN_KEY_CHECKS = 1;

ALTER TABLE transaction ADD CONSTRAINT transaction_idfk FOREIGN KEY (credit_card_id) REFERENCES credit_card(id);
```

Con la creación de la nueva tabla "credit_card" se crea una relación de 1:N a la tabla "transaction". También se ha modificado el campo "expiring_date" con un ALTER TABLE STR_TO_DATE() para que el campo sea en formato fecha YYYY-MM-DD y favorecer el filtrado en los ejercicios siguientes.

Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

The screenshot displays a database management interface with two panels. The left panel shows a SQL script with the following queries:

```
98 -----
99 • SELECT *
100 FROM credit_card
101 WHERE id = 'CcU-2938';
102 • UPDATE credit_card
103 SET iban = 'TR323456312213576817699999'
104 WHERE id = 'CcU-2938';
105 • SELECT *
106 FROM credit_card
107 WHERE id = 'CcU-2938';
```

Below the script, the 'Result Grid' shows the results of the queries:

id	iban	pin	cvv	expiring_date
CcU-2938	TR301950312213576817638661	3257	984	2022-10-30

The right panel shows the same SQL script, but with the results of the UPDATE query displayed in the 'Action Output' section:

#	Time	Action	Message
1	15:01:41	SELECT * FROM transactions.credit_card	5001 row(s) returned
2	15:02:45	SELECT * FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned

EJERCICIO 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

--- EJERCICIOS 6

--- En la taula transaction ingressa una nova transacció amb la següent informació:

```
INSERT INTO transactions.company (id, company_name, phone, email, country, website)
VALUES ('b-9999', "Test Empresa", "123456789", "test_empresa@test.com", "España", "test_empresa.com");
INSERT INTO transactions.credit_card (id, iban, pan, pin, cvv, expiring_date)
VALUES ('CcU-9999', 'XX000000000000000000000000', '0000000000000000', '0123', '123', '2026-09-08');
INSERT INTO transactions.transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined)
VALUES ('10881D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, null, 111.11, 0);
```

```

5 • select * from transactions.transaction
6   where company_id = 'b-9999';
7

```

Result Grid									
Filter Rows:									
Edit:   									
Export/Import:  									
Wrap Cell Content: ☐									
	id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
▶	10881D1D-5B23-A76C-55EF-C568E49A99DD	CcU-9999	b-9999	9999	829.999	-117.999	NULL	111.11	0
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Para completar este ejercicio, primero se debía crear en las tablas “company” y “credit_card” las entradas correspondientes, ya que “transaction” tiene 2 FK (una de cada tabla), por lo que el INSERT INTO daba error.

Con los valores que se tenían que introducir se ha creado una tarjeta de crédito y empresa ficticia a partir de los campos “credit_card_id” y “company_id”.

EJERCICIO 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.

```

112 • ALTER TABLE credit_card DROP COLUMN pan;
113 • SELECT *
114   FROM credit_card
115  LIMIT 10;
116
117
118

```

Result Grid									
Filter Rows:									
Edit:   									
Export/Import:  									
Wrap Cell Content: ☐									
	id	iban	pin	cvv	expiring_date				
▶	CcS-4857	XX4857591835292505850771	1819	467	2025-09-27				
	CcS-4858	XX8581768137002436094025	3964	817	2028-12-28				
	CcS-4859	XX7826930491423553609370	4983	277	2026-11-26				
	CcS-4860	XX5559590368835304645299	6876	661	2027-07-27				
	CcS-4861	XX2035182877195191627307	5710	398	2026-04-25				
	CcS-4862	XX4774721462463645409758	4042	174	2026-11-27				
	CcS-4863	XX1476829664245046207111	5969	449	2029-12-27				
	CcS-4864	XX8380298893385731196159	8481	139	2026-02-28				

credit_card 2 x

Output

Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american_users.csv
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

La taula de products.csv l'utilitzarem més endavant.

```

-----
--- USAR OTRO ESQUEMA
CREATE SCHEMA IF NOT EXISTS business_analytics;
USE business_analytics;
--- Modificar la conexión para poder cargar datos csv (subida bloqueada)
SHOW GLOBAL VARIABLES LIKE 'local_infile';
SET GLOBAL local_infile = ON;

```

Antes de hacer la carga es necesario modificar la conexión. MySQL bloquea por seguridad la carga de datos. Agregar OPT_LOCAL_INFILE=1 en configuración avanzada a la conexión del servidor. También se ha creado otro esquema para este ejercicio llamado “business_analytics”.

Para cargar archivos CSV con LOAD DATA primero hay que crear las tablas. Recomendable inspeccionar primero los valores de los csv. ** Ejemplo con “american_users” **

```

156 • DROP TABLE IF EXISTS american_users;
157 • CREATE TABLE IF NOT EXISTS american_users (
158   id INT,
159   name VARCHAR(50),
160   surname VARCHAR(50),
161   phone VARCHAR(50),
162   email VARCHAR(50),
163   birth_date VARCHAR(25),
164   country VARCHAR(25),
165   city VARCHAR(25),
166   postal_code VARCHAR(25),
167   address VARCHAR(50),
168   signup_date DATE,
169   user_segment VARCHAR(50),
170   income_band VARCHAR(25),
171   region VARCHAR(10));
172

```

```

---
149 --- Cargar la primera tabla (americans_user.csv):
150 • LOAD DATA LOCAL
151   INFILE "C:/Users/noel/Documents/GitHub/IT_Data/Especializacion/2_Sprint/N1-Ex.8__american_users.csv"
152   INTO TABLE american_users
153   FIELDS TERMINATED BY ','
154   ENCLOSED BY '"'
155   IGNORE 1 ROWS;
156
157 --- Comprobación
158 • SELECT *
159   FROM american_users;
160
161 --- Resto de tablas y su carga
162 --- european users

```

id	name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment
1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-78 18 Sagitts St.	2014-04-23	young_professiona
2	Garrett	Mcconnell	(718) 257-2412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave	2014-07-04	young_professiona
3	Ciaran	Harrison	(522) 598-1365	interdum.feugiat@aol.org	Apr 29, 1998	United States	Houston	77001	736-2063 Tellus St.	2014-02-15	frequent_shopper
4	Howard	Stafford	1-411-740-3269	omare.egestas@icloud.edu	Feb 18, 1989	United States	Phoenix	85001	Ap #545-2244 Erat. Rd.	2014-08-21	premium
5	Hayfa	Pierce	1-554-541-2077	eT.malesuada.fames@hotmail.org	Sep 26, 1998	United States	Philadelphia	19101	341-2821 Ultrices Av.	2013-04-28	frequent_shopper
6	Joel	Tyson	(718) 288-8020	gravida.nunc.sed@yahoo.ca	Oct 15, 1989	United States	San Jose	95101	888-2799 Amet Street	2013-08-16	young_professiona
7	Rafael	Jimenez	(817) 689-0478	eget@outlook.ca	Dec 4, 1981	United States	Chicago	60601	8627 Malesuada Rd.	2013-01-15	frequent_shopper

Una vez creadas las tablas, se unen “american_users” y “european_users” al tener la misma estructura y tener la información relacionada con usuarios. Además, se crea una columna “Region” con los valores de “Europa” y “America” para no perder información y facilitar un posible filtrado futuro según la región que sea. También se cambia el formato de “expiring_date” de la tabla credit_cards a formato YYYY-MM-DD. Completados los cambios, se procede a relacionarlas. Era posible hacerlo antes, pero también podría dar error si los datos no están normalizados.

```

CREATE TABLE IF NOT EXISTS users AS
SELECT * FROM american_users
UNION
SELECT * FROM european_users;

```

- Primero, se crean las PK en las tablas:

```

--- Relacion de tablas
--- PK:
ALTER TABLE users ADD CONSTRAINT pk_users_id PRIMARY KEY (id);
ALTER TABLE companies ADD CONSTRAINT pk_companies_id PRIMARY KEY (company_id);
ALTER TABLE credit_cards ADD CONSTRAINT pk_credit_cards_id PRIMARY KEY (id);
ALTER TABLE transactions ADD CONSTRAINT pk_transactions_id PRIMARY KEY (id);

```

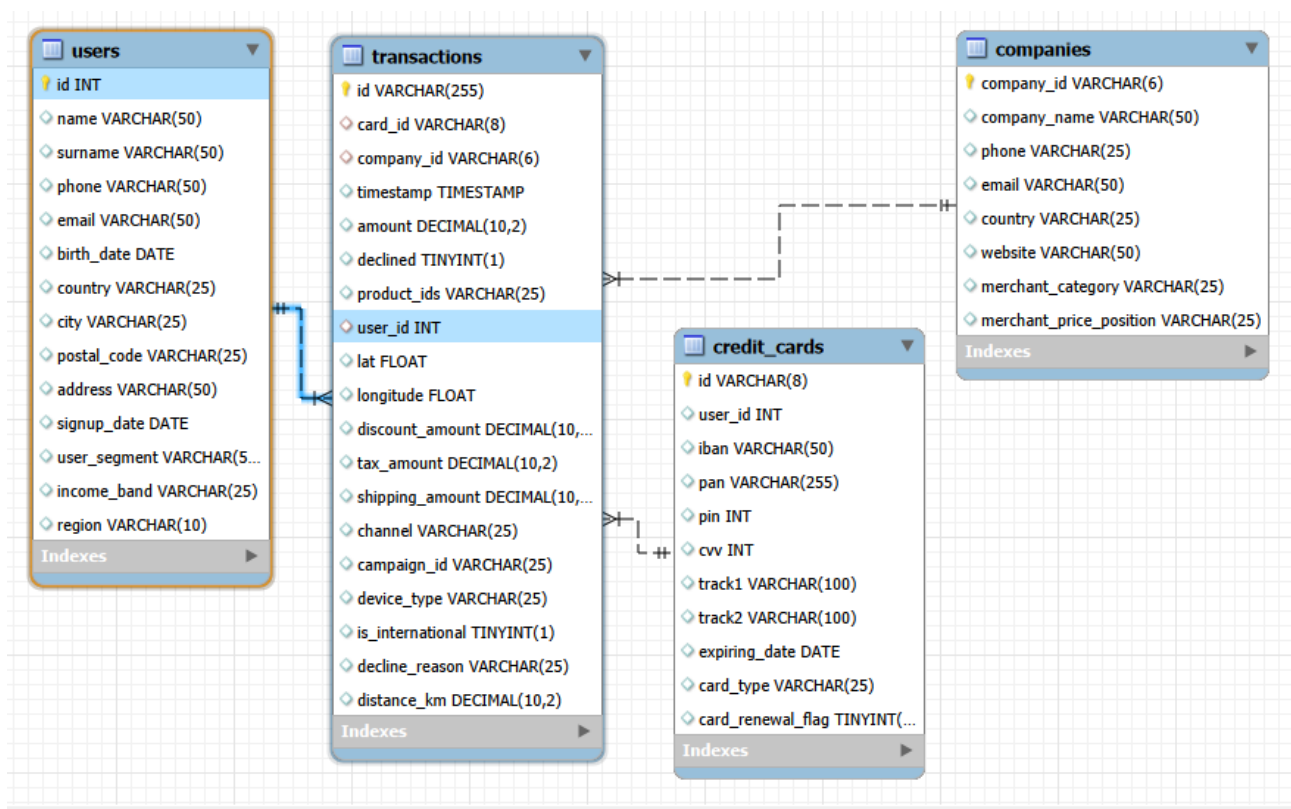
- Segundo, crear las FK:

```

--- FK:
ALTER TABLE transactions ADD CONSTRAINT fk_transactions_cardid FOREIGN KEY (card_id) REFERENCES credit_cards(id);
ALTER TABLE transactions ADD CONSTRAINT fk_transactions_userid FOREIGN KEY (user_id) REFERENCES users(id);
ALTER TABLE transactions ADD CONSTRAINT fk_transactions_companyid FOREIGN KEY (company_id) REFERENCES companies(company_id);

```

El resultado final es un modelo tipo estrella con la tabla “transactions” como tabla de hecho y “credit_cards”, “users” y “companies” como tablas de dimensiones.



Las tablas de dimensión (“users”, “companies” y “credit_cards”) tienen una relación de 1:N con la tabla de hechos (“transactions”).

EJERCICIO 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```
314 -----
315 --- EJERCICIOS 9
316 --- Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.
317 -----
318 • SELECT u.*, subq.n_transactions
319 FROM users as u
320 INNER JOIN
321     (SELECT user_id, count(id) as 'n_transactions'
322      FROM transactions
323      GROUP BY user_id) as subq
324 ON u.id = subq.user_id
325 WHERE subq.n_transactions > 80;
326 -----
```

id	name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment	income_band	n_transactions
185	Molly	Gillam	0800 120 8023	donec@outlook.couk	1993-12-21	United Kingdom	London	EC1A 1BB	P.O. Box 202, 5638 Mi Rd.	2013-08-11	frequent_shopper	high	110
289	Dxwgi	Hwrcu	+98-309-8797	dxwgi.hwrcu@example.com	1976-08-20	Germany	Stuttgart	70173	82 Hwrcu Street	2013-12-11	occasional	low	94
318	Bnyr	Astuw	+33-120-9644	bnyr.astuw@example.com	1974-05-03	Italy	Genoa	16100	53 Astuw Street	2013-05-29	frequent_shopper	very_high	91
454	Sfzzoh	Xgvfrdxs	+58-495-3945	sfzzoh.xgvfrdxs@example.com	1962-08-28	Poland	Gdansk	80-001	52 Xgvfrdxs Street	2014-08-30	established	medium	81

EJERCICIO 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
327 --- EJERCICIOS 10
328 --- Mostra la mitjana d amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.
329 -----
330 • SELECT cc.iban, avg(t.amount) as 'media_amount', count(t.id) as 'n_transactions', c.company_name
331 FROM credit_cards as cc
332 JOIN transactions as t
333 ON cc.id = t.card_id
334 JOIN companies as c
335 ON t.company_id = c.company_id
336 WHERE c.company_name = 'Donec Ltd'
337 GROUP BY cc.iban;
338
339
340
```

iban	media_amount	n_transactions	company_name
CA901355773586771626108990	361.010000	3	Donec Ltd
U5729212642816955651428791	142.560000	1	Donec Ltd
U5899450803230933276819251	246.900000	1	Donec Ltd
U5581097432668325541142297	154.010000	1	Donec Ltd
CA441226267105857473570574	250.510000	1	Donec Ltd
CA332171962352073015086775	203.950000	1	Donec Ltd
U5985359628701198880628720	438.160000	1	Donec Ltd
U5788111972997693675524914	520.366667	3	Donec Ltd
CA867886529258577442049898	204.630000	1	Donec Ltd
U5450950742400388023162002	165.075000	2	Donec Ltd
U5841589234795722857348646	253.930000	1	Donec Ltd
CA707350503853933968563487	290.570000	1	Donec Ltd
U5225174296147002265631190	240.320000	1	Donec Ltd
CA818325250352269640720088	86.520000	1	Donec Ltd
U5570082429286901600591866	288.190000	1	Donec Ltd
U5183391626605377230916689	83.540000	1	Donec Ltd
U5359617522425178629391841	96.805000	2	Donec Ltd
U5494865682709580479552703	158.250000	1	Donec Ltd
CA189943741363518302698531	308.120000	1	Donec Ltd

Result 6 x

#	Time	Action	Message
1	12:33:41	SELECT cc.iban, avg(t.amount) as 'media_amount', count(t.id) as 'n_transactions', c.company_name FROM credit_cards as cc JOIN transactions as t ...	371 row(s) returned

NIVEL 2

EJERCICIO 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
389 SELECT c.company_name, sub1.fecha, sub1.total_amount, sub1.puesto
390 FROM companies as c
391 JOIN
392 (SELECT t.fecha, (sum(t.amount)-sum(t.discount_amount)) as 'total_amount', c.company_id,
393 ROW_NUMBER() OVER (PARTITION BY c.company_name ORDER BY (sum(t.amount)-sum(t.discount_amount)) DESC) AS puesto
394 FROM transactions as t
395 JOIN companies as c
396 ON t.company_id = c.company_id
397 WHERE t.declined <> 1
398 GROUP BY t.fecha,c.company_id
399 ORDER BY c.company_id, puesto) as sub1
400 ON c.company_id = sub1.company_id
401 WHERE puesto <= 5
402 ORDER BY c.company_name, sub1.puesto;
```

Result Grid

	company_name	fecha	total_amount	puesto
	A Institute	2021-09-25	1274.68	1
▶	A Institute	2016-12-30	950.11	2
	A Institute	2019-07-01	896.50	3
	A Institute	2016-02-26	814.93	4
	A Institute	2019-12-06	813.46	5

Result 1

Output

Action Output

#	Time	Action	Message
✓	11 16:14:36	UPDATE transactions SET hora = TIME(timestamp)	100000 row(s) affected Rows matched: 100000 Changed: 100000 Warnings: 0
✓	12 16:14:46	ALTER TABLE transactions DROP COLUMN timestamp	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✗	13 16:15:12	SELECT c.company_name, sub1.fecha, sub1.total_amount, sub1.puesto FROM companie...	Error Code: 1054. Unknown column 'cc.declined' in 'where clause'
✓	14 16:15:24	SELECT c.company_name, sub1.fecha, sub1.total_amount, sub1.puesto FROM companie...	500 row(s) returned

EJERCICIO 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
381 SELECT c.company_name, c.phone, c.country, t.fecha, t.amount
382 FROM companies as c
383 JOIN transactions as t
384 ON c.company_id = t.company_id
385 WHERE t.fecha IN ('2015-04-29', '2018-07-20', '2024-03-13')
386 ORDER BY t.amount DESC;
```

Result Grid

	company_name	phone	country	fecha	amount
▶	Non Institute	+15 722 768 6408	United Kingdom	2018-07-20	695.42
	Egestas Nunc Sed Limited	+39 297 745 2202	Italy	2015-04-29	690.68
	Sed Id Limited	+26 903 356 2276	United States	2015-04-29	674.54
	Sed Nunc Ltd	+5 218 900 3193	United Kingdom	2018-07-20	640.85
	Non Justo Corp.	+67 963 294 1666	Sweden	2018-07-20	604.78
	Maecenas Malesuada Fringilla Inc.	+51 673 540 5289	Netherlands	2018-07-20	569.86
	Integer Mollis Corp.	+38 803 863 7881	Italy	2024-03-13	567.60
	Tristique Neque Venenatis Institute	+18 605 905 7882	Sweden	2018-07-20	543.46
	Mauris Incorporated	+26 419 837 1876	Norway	2018-07-20	493.02

Result 14

EJERCICIO 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
395 • SELECT c.company_name,  
396 CASE  
397 WHEN count(t.id) >= 400 THEN 'Igual o más de 400 transacciones'  
398 ELSE 'Menos de 400'  
399 END as 'total_transactions'  
400 FROM companies as c  
401 JOIN transactions as t  
402 ON c.company_id = t.company_id  
403 GROUP BY c.company_name  
404 ORDER BY c.company_name;  
405  
406  
407  
408
```

Result Grid	
Filter Rows:	Export: Wrap Cell Content:
company_name	total_transactions
A Institute	Igual o más de 400 transacciones
Ac Fermentum Incorporated	Igual o más de 400 transacciones
Ac Industries	Igual o más de 400 transacciones
Ac Libero Inc.	Igual o más de 400 transacciones
Aliquam Erat Volutpat LLP	Igual o más de 400 transacciones
Aliquam Taculis Lacus Corp.	Igual o más de 400 transacciones
Aliquam PC	Igual o más de 400 transacciones
Aliquet Diam Limited	Igual o más de 400 transacciones
Aliquet Sem Limited	Igual o más de 400 transacciones

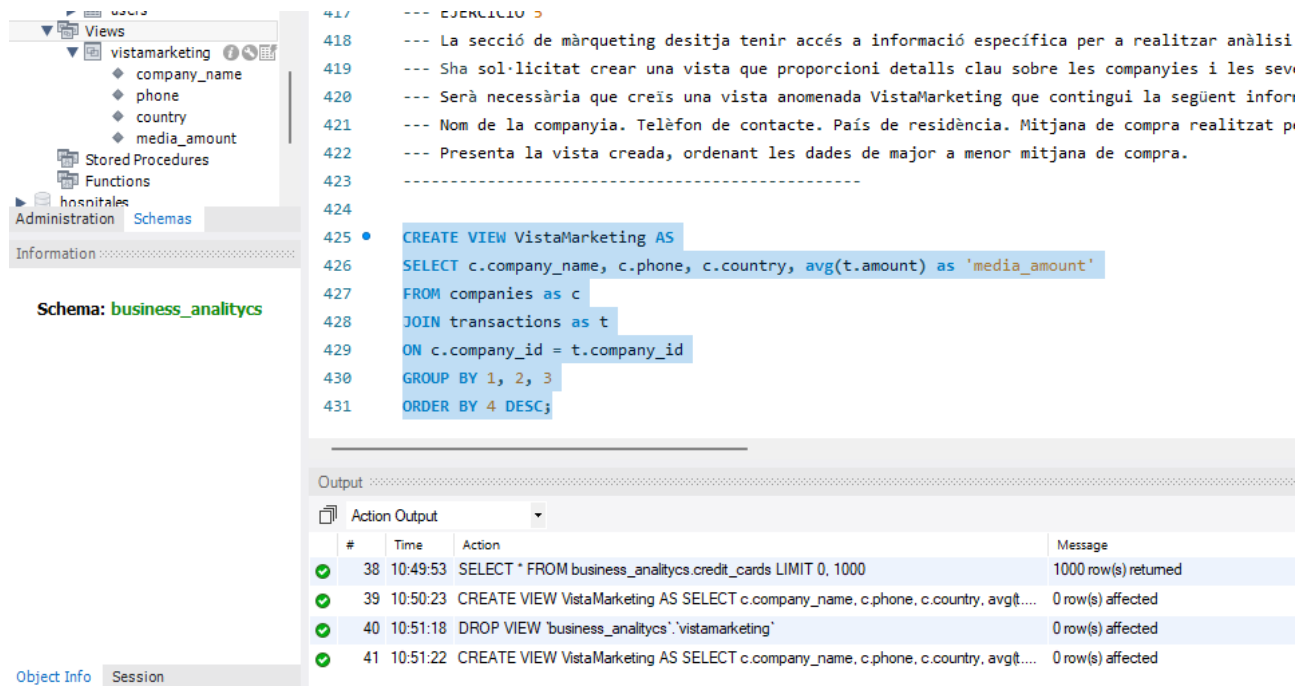
EJERCICIO 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```
411 • SELECT count(*)  
412 FROM transactions;  
413 • DELETE FROM transactions  
414 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';  
415
```

EJERCICIO 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.



```
417  --- EJERCICIO 5
418  --- La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi
419  --- S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves
420  --- Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació:
421  --- Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per
422  --- Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.
423  -----
424
425  CREATE VIEW VistaMarketing AS
426  SELECT c.company_name, c.phone, c.country, avg(t.amount) as 'media_amount'
427  FROM companies as c
428  JOIN transactions as t
429  ON c.company_id = t.company_id
430  GROUP BY 1, 2, 3
431  ORDER BY 4 DESC;
```

Schema: business_analyticscs

Output

#	Time	Action	Message
✓ 38	10:49:53	SELECT * FROM business_analyticscs.credit_cards LIMIT 0, 1000	1000 row(s) returned
✓ 39	10:50:23	CREATE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.country, avg(t....	0 row(s) affected
✓ 40	10:51:18	DROP VIEW 'business_analyticscs'.vistamarketing	0 row(s) affected
✓ 41	10:51:22	CREATE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.country, avg(t....	0 row(s) affected

NIVEL 3

EJERCICIO 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:

```
474 • CREATE TABLE IF NOT EXISTS credit_card_stat
475     SELECT card_id,
476     CASE
477         WHEN sum(declined)=3 THEN 'Inactiva'
478         ELSE 'Activa'
479     END as stat
480     FROM
481     (SELECT t.card_id, t.fecha, t.declined,
482     ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY fecha DESC) AS puesto
483     FROM transactions as t
484     JOIN companies as c
485     ON t.company_id = c.company_id) as sub1
486     WHERE puesto < 3
487     GROUP BY card_id
```

👉 Quantes targetes estan actives?

```
490 --- Quantes targetes estan actives? 5000 No hay Inactivas
491 • SELECT count(card_id)
492 FROM credit_cards_stat
493 WHERE stat = 'Activa';
494
495
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
count(card_id)			
5000			

5000 targetes actives. No hay targetes inactivas.

EJERCICIO 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

Primero cargamos la tabla “productos” y cargamos los datos:

```
519 • DROP TABLE IF EXISTS productos;
520 • CREATE TABLE IF NOT EXISTS productos (
521   id INT,
522   product_name VARCHAR(50),
523   price VARCHAR(15),
524   colour VARCHAR(15),
525   weight DECIMAL(10,1),
526   warehouse_id VARCHAR(8),
527   category VARCHAR(15),
528   brand VARCHAR(15),
529   cost VARCHAR(15),
530   launch_date DATE);
531
532 • LOAD DATA LOCAL
533   INFILE "C:/Users/noel_/Documents/GitHub/IT_Data/Especializacion/2_Sprint/N1-Ex.8__products.csv"
534   INTO TABLE productos
535   FIELDS TERMINATED BY ','
536   ENCLOSED BY ''
```

Output

Action Output

#	Time	Action	Message
5	15:06:21	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	Error Code: 1146. Table 'transactions.product_transaction' doesn't exist
6	15:06:26	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	100 row(s) returned
7	15:07:00	DROP TABLE IF EXISTS productos	0 row(s) affected
8	15:07:00	CREATE TABLE IF NOT EXISTS productos (id INT, product_name VARCHAR(50), price ...	0 row(s) affected

```
532 • LOAD DATA LOCAL
533   INFILE "C:/Users/noel_/Documents/GitHub/IT_Data/Especializacion/2_Sprint/N1-Ex.8__products.csv"
534   INTO TABLE productos
535   FIELDS TERMINATED BY ','
536   ENCLOSED BY ''
537   IGNORE 1 ROWS;
538
539 • CREATE TABLE IF NOT EXISTS product_transaction
540   SELECT t.id as transaction_id, p.id as product_id, t.product_ids as transactions_products_ids, t.fecha
541   from productos as p
542   JOIN transactions as t
```

Output

Action Output

#	Time	Action	Message	Dt
6	15:06:26	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	100 row(s) returned	0.5
7	15:07:00	DROP TABLE IF EXISTS productos	0 row(s) affected	0.1
8	15:07:00	CREATE TABLE IF NOT EXISTS productos (id INT, product_name VARCHAR(50), price ...	0 row(s) affected	0.0
9	15:07:44	LOAD DATA LOCAL INFILE "C:/Users/noel_/Documents/GitHub/IT_Data/Especializaci...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.0

Creamos una tabla puente de productos-transacciones llamada: product_transaction:

```

539 • DROP TABLE IF EXISTS product_transaction;
540 • CREATE TABLE IF NOT EXISTS product_transaction
541 SELECT t.id as transaction_id, p.id as product_id, t.product_ids as transactions_products_ids, t.fecha
542 from productos as p
543 JOIN transactions as t
544 ON FIND_IN_SET(p.id, REPLACE(t.product_ids, ' ', '')) -- product_ids es un conjunto de ids por lo que no vale el p.id in t.product_ids
545 ORDER BY product_ids, p.id;
546
547 --- SOLUCION FINAL
548 • SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas'
549 FROM product_transaction
550 GROUP BY product_id
551 ORDER BY 2 DESC
552

```

Output				
Action Output				
#	Time	Action	Message	Duration
✓ 11	15:08:43	CREATE TABLE IF NOT EXISTS product_transaction SELECT t.id as transaction_id, p.id ...	2654 row(s) affected Records: 2654 Duplicates: 0 Warnings: 0	1.203
✓ 12	15:11:32	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	1 row(s) returned	0.016
✓ 13	15:11:48	DROP TABLE IF EXISTS product_transaction	0 row(s) affected	0.328
✓ 14	15:11:48	CREATE TABLE IF NOT EXISTS product_transaction SELECT t.id as transaction_id, p.id ...	253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0	12.64

Se utiliza FIND_IN_SET para buscar el id del producto en los ids que hay en la tabla “transactions”. Al estar agrupados en un mismo campo (un varchar) y no una lista, hay que reemplazar los espacios por nada y de esta manera es capaz de encontrarlos.

👉 **Necessitem conèixer el nombre de vegades que s'ha venut cada producte.**

```

547 --- SOLUCION FINAL
548 • SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas'
549 FROM product_transaction
550 GROUP BY product_id
551 ORDER BY 2 DESC
552

```

product_id	n_ventas
52	2654
29	2635
21	2609
16	2608
66	2600

Output				
Action Output				
#	Time	Action	Message	Duration / Fet
✓ 12	15:11:32	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	1 row(s) returned	0.016 sec / 0
✓ 13	15:11:48	DROP TABLE IF EXISTS product_transaction	0 row(s) affected	0.328 sec
✓ 14	15:11:48	CREATE TABLE IF NOT EXISTS product_transaction SELECT t.id as transaction_id, p.id ...	253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0	12.641 sec
✓ 15	15:12:30	SELECT distinct(product_id) as product_id, count(product_id) as 'n_ventas' FROM product...	100 row(s) returned	0.391 sec / 0